

Classification-Based Approach to Concept Map Generation in Adaptive Learning

Xiaopeng Huang

Smilek12, Inc.

Freehold, NJ 07728, United States

e-mail: xhuang@smilek12.com

Kyeong Yang and Victor Lawrence

Center for Intelligent Networked Systems

Stevens Institute of Technology

Hoboken, NJ 07030, United States

e-mail: {kyeong.yang, victor.lawrence}@stevens.edu

Abstract—Data mining has recently drawn a lot of interests as an effective way of generating a concept map in an adaptive learning system that provides students with the personalized learning guidance. Even with significant progresses witnessed in this field, the data mining-based concept map generation needs further improvement both in accuracy and complexity before it can be employed in actual education services. This paper proposes a classification-based approach to significantly reduce computational complexity of concept map generation while maintaining the accuracy of the generated concept map, and demonstrates through simulations that the approach accomplishes the objectives.

Keywords—data mining; concept map generation; Apriori algorithm; classification process

I. INTRODUCTION

With recent advancement in data mining-based adaptive learning techniques, there have been proposed several adaptive e-learning systems that offer students customized course content in accordance with their aptitudes and learning results [1-3]. Adaptive learning usually involves adaptive content aggregation, adaptive recommendation, adaptive presentation, adaptive navigation (concept map), and student knowledge score calculation in order to provide students with the best learning experience.

A concept map consists of key concepts and links that illustrate relationships among those concepts, and plays a critical role in providing adaptive learning guidance to students and evaluating students' knowledge level of various concepts, among others. However, it is difficult and time consuming to manually create a concept map for a courseware. Thus, there have been efforts to automatically create a concept map. *Tseng et al* developed the two-phase concept map generation (TP-CMC) approach to automatic generation of concept maps by using students' historical test records [4]. *Bai et al* presented a concept map generation method based on fuzzy rules and students' testing records for adaptive learning systems [5]. *Lee et al* proposed an approach that utilizes the Apriori algorithm, and developed the remedial-instruction path (RIP) algorithm to build an intelligent concept diagnostic system [6].

Key metrics for quality of concept map generation algorithms are their accuracy and complexity. The accuracy is critical in concept maps, but most papers did not evaluate the accuracy of their proposed algorithms comparing them with others mainly because there is no standard way and

courseware commonly used in evaluation. However, close reviews of the papers teach that there is a room for improvement of accuracy by adopting more appropriate formulas. While the complexity of existing concept map generation algorithms seems to be low with data of a small scale, the complexity dramatically increases with an increased amount of data to be processed [7, 8].

This paper proposes an efficient data mining approach to concept map generation that employs classification of test records to significantly reduce the complexity while maintaining the accuracy. Section II describes the proposed algorithm and analyzes the complexity of the conventional concept map generation approach and the proposed approach, and Section III presents an example of the proposed concept map generation and discusses the complexity of the algorithm. Section IV concludes the paper.

II. ALGORITHM DESCRIPTION

A. Overview of the proposed algorithm

Fig. 1 depicts the proposed concept map generation algorithm that introduces classification of student test records to reduce computational complexity before applying the data mining process. As shown in the figure, classification is performed on student test records according to a pre-determined rule on student learning data (each student's knowledge level) and courseware data (difficulty level of each question), and data mining is performed on each subset to calculate relevance degrees among concepts for the respective subset. Then, results for all subsets selected are merged to obtain resultant relevance degrees among concepts that form a concept map together.

The proposed data mining process, shown in the dashed block in Fig. 1, deals with the wrong-to-wrong relationship of questions for Subset k . This process adopts the Apriori algorithm [9-10] to perform data mining for concept map generation. It first performs *largest itemset mining* to reduce the number of association rules with a little degradation in accuracy, and then *association rule and confidence mining* to find survived rules used in concept map generation and calculate their confidence (probability) values. Contribution of each selected rule to the relevance degree between two concepts is calculated to connect student test records to question-to-concept relationship data, and the contributions are added up to calculate relevance degrees among concepts

for Subset k . Courseware data is used in calculating the contribution of each rule to concept relevance degree.

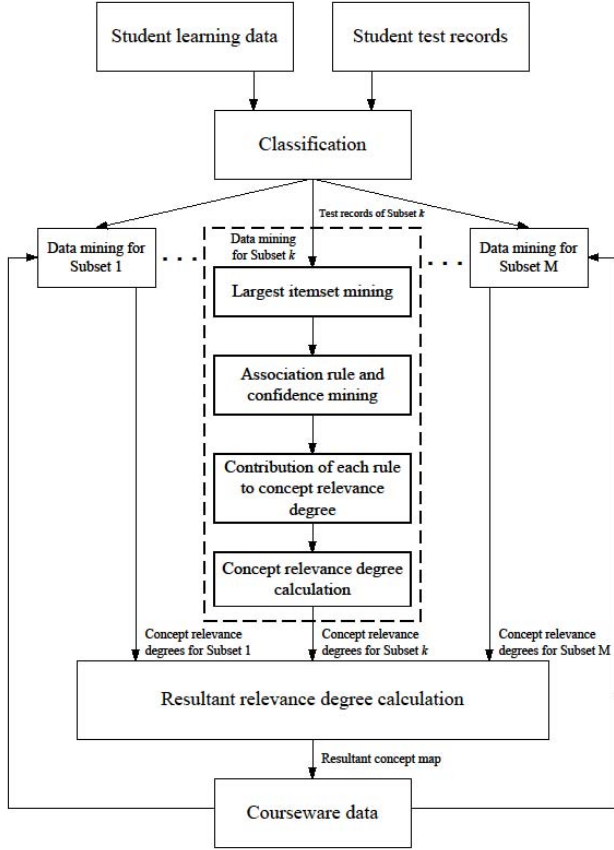


Figure 1. The proposed concept map generation algorithm employing classification of learning information.

The classification process divides student records into multiple subsets according to a pre-determined criterion. Fig. 2 is an example that illustrates a classification rule of student test records. We divide questions into N_{subset_q} levels in terms of the difficulty level, and students into N_{subset_s} levels in terms of their knowledge level. Therefore, the total number of subsets is calculated as in Eq. (1):

$$N_{subset} = N_{subset_s} \times N_{subset_q} \quad (1)$$

In the example classification rule shown in Fig. 2, the student test record is divided into 9 subsets, and then the classification function selects M subsets according to the rule that is detailed in Section II.C.

B. Definitions for test records

Below lists notations of parameters used in description of the proposed algorithm:

- N_Q : total number of questions
- N_S : total number of students
- N_R : number of rules survived

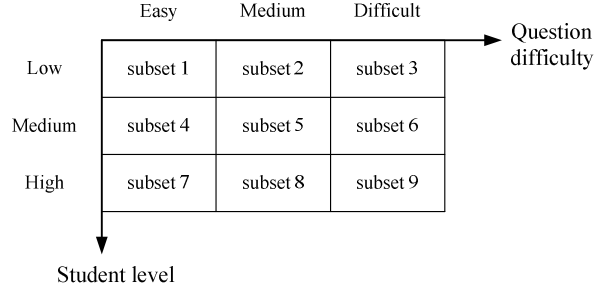


Figure 2. An example rule for classification of student test records.

- N_C : number of concepts that have anything to do with the questions associated with the survived rules.
- $N_Q(k)$: number of questions in Subset k
- $N_S(k)$: number of students in Subset k
- $N_R(k)$: number of association rules survived in Subset k
- $N_C(k)$: number of concepts that have anything to do with questions associated with the survived rules in Subset k .
- $R_{C_x C_y}$: concept relevance degree from C_x to C_y .
- P_w : probability of wrong answers in total answers
- $sup(Q_{ik})$: support value of the 1-itemset Q_i in Subset k
- $sup(Q_{ik}, Q_{jk})$: support value of the 2-itemset (Q_i, Q_j) in Subset k

The proposed algorithm processes two types of data for concept map generation: one is student test record (e.g., student test result for questions and student levels), and the other is relationship of questions to various concepts that may be generated by experienced teachers and obtained from the courseware. Student test record is practical and dynamic data, which is processed by data mining algorithms such as clustering, classification, and association rule.

The grade of a question is generally in the range of $[0, R]$, where R is an integral number larger than 0, and can be classified into N clusters. In this paper, we choose R to be 1 so that the grade has only two values, 0 and 1, 0 for the correct answer, and 1 for the wrong answer.

We also define matrixes for student test results (grade for each question) and relationships of questions to various concepts. Assume there are N_s students ($S_1, S_2, \dots, S_{N_s}$) and N_Q questions ($Q_1, Q_2, \dots, Q_{N_Q}$) in the test record. Then, the matrix for student test results is defined as in Eq. (2):

$$M_{QS} = \begin{matrix} & Q_1 & Q_2 & \cdots & Q_{N_Q} \\ \begin{matrix} S_1 \\ S_2 \\ \vdots \\ S_{N_s} \end{matrix} & \begin{bmatrix} qs_{11} & qs_{12} & \cdots & qs_{1N_Q} \\ qs_{21} & qs_{22} & \cdots & qs_{2N_Q} \\ \vdots & \vdots & \vdots & \vdots \\ qs_{N_s 1} & qs_{N_s 2} & \cdots & qs_{N_s N_Q} \end{bmatrix} \end{matrix} \quad (2)$$

where qs_{ij} denotes the grade of the i^{th} student S_i for the j^{th} question Q_j , and $qs_{ij} \in [0, R]$, $1 \leq i \leq N_s$, and $1 \leq j \leq N_Q$.

For the case of N_C concepts ($C_1, C_2, \dots, C_{N_C}$) involved in the test, we define a question-to-concept relationship matrix as in Eq. (3):

$$M_{QC} = \begin{matrix} & C_1 & C_2 & \dots & C_{N_C} \\ \begin{matrix} Q_1 \\ Q_2 \\ \vdots \\ Q_{N_Q} \end{matrix} & \begin{bmatrix} qc_{11} & qc_{12} & \dots & qc_{1N_C} \\ qc_{21} & qc_{22} & \dots & qc_{2N_C} \\ \vdots & \vdots & \vdots & \vdots \\ qc_{N_Q1} & qc_{N_Q2} & \dots & qc_{N_QN_C} \end{bmatrix} \end{matrix} \quad (3)$$

where qc_{ij} denotes the relevance score of the i^{th} question Q_i to the j^{th} concept C_j , $qc_{ij} \in [0, R_S]$, $1 \leq i \leq N_Q$, and $1 \leq j \leq N_C$. R_S is the range of relevance score, and $qc_{ij} = 0$ means that the i^{th} question Q_i does not have any relation with the j^{th} concept C_j .

C. Classification and selection Rule

The proposed algorithm classifies the student test record into multiple subsets according to a pre-determined rule, and selects M subsets that are considered to better represent the relationship of various concepts. In this paper, we equally distribute students and questions across all subsets, and select subsets based on the probability of the wrong answer of student data belonging to the subset, which is denoted as p_w . That is, we select subsets with $p_{low} \leq p_w \leq p_{high}$ considering that if the probability, p_w , is too low (or too high), almost everybody answered it correctly (or incorrectly) and therefore there will be only a little information we can exploit. We set p_{low} and p_{high} respectively to 0.3 and 0.5 in our simulations.

D. Algorithm description

The proposed algorithm is run with the student test record and relevance scores of each question to various concepts that are stored in the courseware, as follows:

Step 1: Classify student test records into N_{subset} subsets using a pre-determined classification rule (e.g., the rule shown in Fig. 2), and select M survived subsets using a pre-determined selection rule (the rule described in Section II. C). We denote the selected subsets as Subset k , $1 \leq k \leq M$, where M is the total number of survived subsets.

Step 2: Perform the largest itemset mining for wrong-to-wrong answers in Subset k to find the most frequent itemset by using an association rule algorithm, such as the Apriori algorithm. Setting minimum support plays a critical role in the largest itemset mining process. The minimum support is used to exclude itemsets whose support values are less than the value. In order to make the largest itemset mining process more effective, the proposed algorithm sets the minimum support based on support values of the 1-itemsets, as in Eq. (4):

$$\text{minimum support} = \left\lceil \frac{\sum \text{sup}(Q_{ik})}{N_Q(k)} \right\rceil - 1 \quad (4)$$

where $\text{sup}(Q_{ik})$ is the support value of Q_i in Subset k , and $\lceil x \rceil$ is the largest integer that doesn't exceed x .

Step 3: Based on the itemsets survived in the largest itemset mining process in Step 2, perform the association rule and confidence mining to generate association rules and their confidence values for Subset k . Confidence is the probability of the event that a student also incorrectly answers the question Q_j when the student answers the question Q_i incorrectly. Therefore, the "confidence ($Q_{ik} \rightarrow Q_{jk}$)" of an association rule " $Q_{ik} \rightarrow Q_{jk}$ " is calculated as in Eq. (5):

$$\text{confidence}(Q_{ik} \rightarrow Q_{jk}) = \frac{\text{sup}(Q_{ik}, Q_{jk})}{\text{sup}(Q_{ik})} \quad (5)$$

where $1 \leq i \leq N_Q$, $1 \leq j \leq N_Q$, and $i \neq j$.

Step 4: Using the relevance of the questions to various concepts provided from the courseware data in the format of the question-to-concept relationship matrix M_{QC} shown in Eq. (2), contribution of the association rule " $Q_{ik} \rightarrow Q_{jk}$ " to the relevance degree of $C_x \rightarrow C_y$ in Subset k , denoted as $R_{xcy_k}(Q_{ik} \rightarrow Q_{jk})$, is calculated in Eq. (6):

$$R_{xcy_k}(Q_{ik} \rightarrow Q_{jk}) = \text{confidence}(Q_{ik} \rightarrow Q_{jk}) \times (R_{Q_i C_x} \cdot R_{Q_j C_y} \cdot R_{Q_i C_y})^{\frac{1}{3}} \quad (6)$$

where $R_{Q_i C_x}$, $R_{Q_j C_y}$, and $R_{Q_i C_y}$ are respectively relevance scores of $Q_i \rightarrow C_x$, $Q_j \rightarrow C_y$, and $Q_i \rightarrow C_y$, that are provided by the courseware in the format of matrix M_{QC} in Eq. (3).

Step 5: Calculate the relevance degree of $C_x \rightarrow C_y$ in Subset k , R_{xcy_k} , as the arithmetic average of relevance degrees of all $N_R(k)$ association rules:

$$R_{C_x C_y_k} = \frac{\sum_{r=1}^{N_R(k)} (d_{rk} \cdot d_{rk})}{\sum_{r=1}^{N_R(k)} d_{rk}} \quad (7)$$

where $d_{rk} = R_{xcy_k}(Q_{ik} \rightarrow Q_{jk})$ denotes the relevance degree for the r^{th} association rule in Subset k .

Step 6: Merge relevance degrees of all selected subsets to calculate the resultant relevance degree among concepts:

$$R_{C_x C_y} = \sum_{k=1}^M \alpha_k R_{C_x C_y_k} \quad (8)$$

where α_k is the weight of Subset k in the merge process.

E. Complexity analysis

The computational complexity is a very important factor when handling test records of a large size. We approximate the complexity with following assumptions:

- Each student answers all questions.
- Consider the complexity of only 1-itemset and 2-itemset in the Apriori algorithm for simplicity. The complexity of higher-order itemsets (e.g., 3-itemset) can be neglected since their complexity highly depends on the number of itemsets survived in the previous itemset mining step and is usually much less.

The complexity of the conventional approach consists of two parts, the complexity of the Apriori algorithm and the complexity of the relevance degree calculation. The complexity for the Apriori algorithm has been analyzed in [9]. The proposed approach includes complexity for the classification process in addition to the two parts.

Table 1 compares the complexity of the proposed algorithm to that of the conventional approach. The order of complexity $O(\cdot)$ represents the arithmetic order of operations which quantify the computational complexity of algorithms.

The number of survived rules N_R can be represented as in Eq. (9) from the definition of the association rule in Step 3 in Section II.D:

$$N_R = N_Q \cdot P_2 \cdot p \quad (9)$$

where p is the probability of rules survived without classification. We may select a lower value for p when N_Q is larger. Therefore, when using classification, we set the probability for Subset k (denoted as p_k) to $p_k = p(1 + \alpha N_{subset})$, where α is a constant (e.g., $\alpha=0.03$).

TABLE I. COMPUTATIONAL COMPLEXITY COMPARISON

Approach	Computational complexity
Conventional approach does not consider classification process	$O(N_Q^2 \cdot N_s) + O[N_R \cdot N_C \cdot P_2]$
Proposed approach applies classification of multiple subsets	$complexity_classification$ $+ \sum_{k=1}^M O[N_Q(k)^2 \cdot N_s(k)]$ $+ \sum_{k=1}^M O[N_R(k) \cdot N_C(k) \cdot P_2]$ $+ complexity_merge$

III. EXAMPLE CONCEPT MAP GENERATION

A. Data collection

In order to demonstrate the proposed approach, we used an example with 12 questions, 4 concepts, and 15 students¹. The difficulty level increases from C_1 to C_4 . Based on the classification model proposed in Fig. 2, the input data matrixes in Eq. (3) and Eq. (2) can be expressed as follows:

$$M_{QC} = \begin{matrix} & Q_1 & Q_2 & Q_3 & Q_4 & Q_5 & Q_6 & Q_7 & Q_8 & Q_9 & Q_{10} & Q_{11} & Q_{12} \\ \begin{matrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{matrix} & \begin{bmatrix} 8 & 9 & 7 & 5 & 4 & 0 & 1 & 0 & 2 & 3 & 0 & 2 \\ 5 & 4 & 5 & 3 & 8 & 3 & 4 & 9 & 1 & 0 & 6 & 4 \\ 1 & 0 & 0 & 2 & 2 & 6 & 7 & 4 & 3 & 9 & 7 & 6 \\ 0 & 1 & 1 & 0 & 0 & 5 & 0 & 1 & 7 & 4 & 4 & 7 \end{bmatrix} \end{matrix} \quad (10)$$

where $[1, 10]$ is the range of relevance score of each question to concept, and 0 between Q_i and C_j means that Q_i does not have any relation with C_j , and

¹ The data was generated by several students and questions in Liner Algebra and Geometry, and has been modified to make it suitable for the purpose of the paper.

$$M_{QS} = \begin{matrix} & \begin{matrix} \text{Easy} & \text{Medium} & \text{Difficult} \end{matrix} & & \begin{matrix} \text{Question} \\ \text{difficulty} \end{matrix} \\ & \begin{matrix} Q_1 & Q_2 & Q_3 & Q_4 & Q_5 & Q_6 & Q_7 & Q_8 & Q_9 & Q_{10} & Q_{11} & Q_{12} \end{matrix} & & \\ \begin{matrix} S_1 \\ S_2 \\ S_3 \\ S_4 \\ S_5 \\ S_6 \\ S_7 \\ S_8 \\ S_9 \\ S_{10} \\ S_{11} \\ S_{12} \\ S_{13} \\ S_{14} \\ S_{15} \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} & \begin{matrix} \text{Low} \\ \\ \\ \\ \\ \\ \\ \text{Medium} \\ \\ \\ \\ \\ \\ \\ \text{High} \end{matrix} \end{matrix} \quad (11)$$

Student level

where one (1) represents the wrong answer, and zero (0) represents the correct answer.

B. Concept map generation

In this example, three subsets are selected through the selection rule described in Section II.C, which are subset 1 (easy questions; low level students), subset 5 (medium questions; medium level students), and subset 9 (difficult questions; high level students), and used in generating a concept map. Fig. 3 shows a concept map generated for subset 5 in terms of wrong-to-wrong answers, and Fig. 4 is the resultant concept map generated by merging results of the 3 subsets with $\alpha_k=1/3$ for all k . Fig. 5 is a concept map generated without using the classification process.

C. Discussions on the concept map

It is worthwhile to mention again that a concept map reveals pre-acquisition relationships between concepts that are used to provide adaptive learning guidance. As mentioned in Section III.A, the difficulty level of concepts increases from C_1 to C_4 meaning that students need to study C_1 before studying C_2 . In Fig. 4, as an example, the relevance degree from C_1 to C_2 is 3.07, and that from C_2 to C_1 is 1.04. This means that C_1 is a pre-acquisition concept of C_2 , and if a student fails in C_1 , it is very probably that he will also fail in C_2 . That is, this shows that the concept map reflects the actual relationship among the concepts we selected.

Through close comparisons of the two concept maps shown in Figs. 4 and 5, respectively for the proposed approach and the conventional approach that doesn't employ classification, we can draw a conclusion that the classification-based concept map maintains the relationship among concepts seen in that from the conventional approach, meaning that the accuracy is not noticeably degraded.

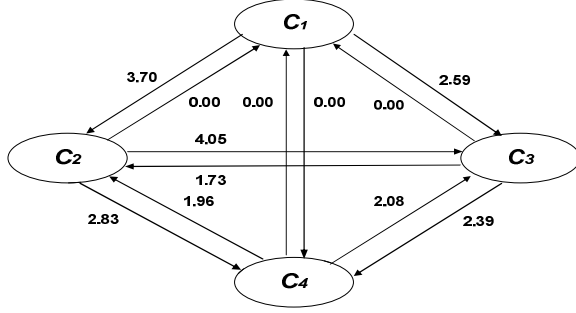


Figure 3. Concept map generated for Subset 5.

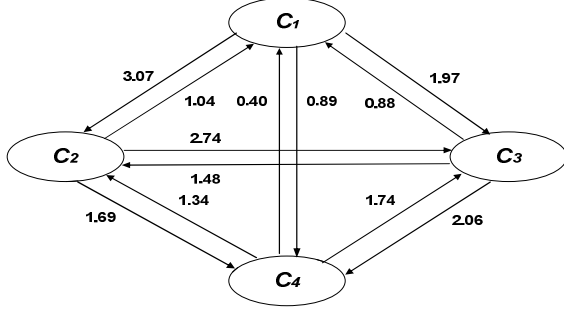


Figure 4. Resultant concept map generated by merging results from three subsets ($\alpha_k = 1/3$).

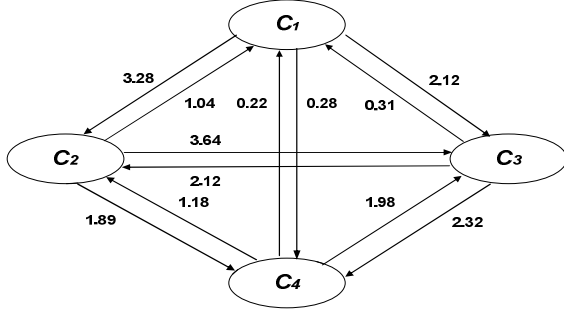


Figure 5. Concept map generated without using the classification process.

D. Complexity evaluation

The evaluation of computational complexity is more meaningful when the concept map generation process involves data of a large size. Therefore, in order to better illustrate complexity reduction of the proposed approach, we propose another example with 300 questions ($N_Q = 300$), 30 students ($N_S = 30$), and 20 concepts 20 ($N_C = 20$), along with the following assumptions:

Assumption 1: The student test data are classified into 9 subsets and 3 subsets are selected by the selection rule.

Assumption 2: We classify questions and students with equal numbers in each subset, and choose $p = 0.2$ in Eq. (9).

Assumption 3: The complexity of merging results from the 3 subsets can be neglected compared to the complexity of other items. The complexity of classification is also negligible because classification of questions and students has already been done in test records and courseware.

Then, the complexity of the conventional approach is:

$$O(2.7 \times 10^6) + O(6.82 \times 10^6) \approx O(9.52 \times 10^6), \quad (12)$$

and that of the proposed approach is:

$$[O(3 \times 2.16 \times 10^4) + O(3 \times 2.97 \times 10^4)] \approx O(3.87 \times 10^5) \quad (13)$$

This shows that our proposed approach significantly reduces the computational complexity.

IV. CONCLUSIONS

In this paper, we proposed a classification-based concept map generation approach that significantly reduces computational complexity while maintaining the accuracy. We also adopted more accurate formulas in calculation of relevance degrees between concepts. We showed through simulations with a practical learning courseware and test data that the proposed approach can generate an accurate and robust concept map with reduced complexity.

REFERENCES

- [1] V. L. Carchiolo and M. Malgeri, "Adaptive formative paths in a web-based learning environment," *Educational Technology and Society*, vol. 5, no. 4, pp. 64-75, 2002.
- [2] J. Yang, Z. X. Huang, Y. X. Gao, and H. T. Liu, "Dynamic learning style prediction method based on a pattern recognition technique," *IEEE Trans. on Learning Technologies*, vol. 7, no. 2, pp. 165-177, 2014.
- [3] A. K. Milicevic, B. Vesin, M. Ivanovic, and Z. Budimac, "E-learning personalization based on hybrid recommendation strategy and learning style identification," *Computers & Education*, vol. 56, pp. 885-899, 2011.
- [4] S. Tseng, P. Sue, J. Su, J. Weng, and W. Tsai, "A new approach for constructing the concept map," *Computers & Education*, vol. 49, pp. 691-707, 2007.
- [5] S. M. Bai and S. M. Chen, "Automatically constructing concept maps based on fuzzy rules for adaptive learning system," *Experts Systems with Applications*, vol. 35, no. 3, pp. 1408-1414, 2008.
- [6] C. H. Lee, G. G. Lee, and Y. H. Leu, "Application of automatically constructed concept map of learning to conceptual diagnosis of e-learning," *Expert Systems with Applications*, vol. 36, no. 2, pp. 1675-1684, 2009.
- [7] S. M. Chen and S. M. Bai, "Using data mining techniques to automatically construct concept maps for adaptive learning systems," *Expert Systems with Applications*, vol. 37, no. 6, pp. 4496-4503, 2010.
- [8] S. M. Chen and P. J. Sue, "Constructing concept maps for adaptive learning systems based on data mining techniques," *Expert Systems with Applications*, vol. 40, no. 6, pp. 2746-2755, 2013.
- [9] M. Hegland, "Mathematics and computation in imaging science and information processing," World Scientific Publisher, 2007.
- [10] X. Wu and V. Kumar, "The top ten algorithms in data mining," Chapman and Hall Publisher, London, 2009.